Assignment 2
Multi-Agent Systems Laboratory
Coordinating the Tumor–Immune Microenvironment

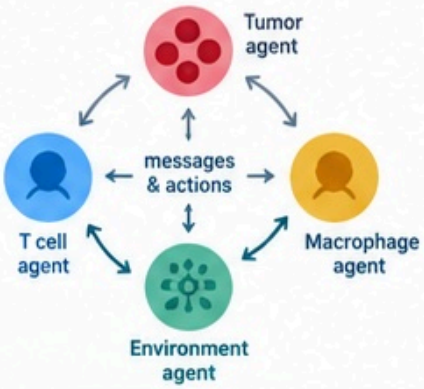
Course 2
Multi-Agent Systems Laboratory

Learning Goal

Design and implement a multi-agent system to coordinate the interactions between tumor cells, immune cells and the tumor microenvironment, based on the model developed in Assignment 1.

Context

The same tumor–immune scenario is now addressed from a multi-agent systems (MAS) perspective. Instead of modelling the system as a monolithic model, each cell type and environmental component is represented by an autonomous agent. Agents interact through explicit communication and coordination mechanisms to achieve local objectives and produce emergent global behaviour.



Assignment Description

Students design, implement and test a multi-agent system that includes:

1. **Agent roles and responsibilities:** tumor cells, cytotoxic T cells, macrophages, and an environment agent (e.g., nutrient/oxygen field).
2. **Agent architecture:** internal state, goals, decision rules and behaviour.
3. **Communication and interaction protocols:** message types, performatives, and coordination mechanisms (e.g., negotiation, signalling, or simple rules).
4. **System integration:** implement the agents using a suitable framework or language (e.g., JADE, SPADE, Mesa, or custom implementation).
5. **Testing and evaluation:** verify correct behaviour, analyse emergent dynamics, and compare with the results of Assignment 1.

Deliverables

- A **technical report** (PDF, max 8 pages) including:
  - System architecture and agent design
  - Communication and coordination protocols
  - Implementation details (tools, libraries, structure)
  - Testing methodology and results
  - Discussion of results, limitations and possible extensions
- A **short live demonstration** (10 minutes) of the running system.

Duration
90 minutes

Team Size
3–5 students

Expected Tools
Any MAS framework or language (e.g., JADE, SPADE, Python, etc.)

Tip
Use the same biological scenario and parameters of Assignment 1 to ensure continuity between the two activities, and to highlight the complementarity between modelling and MAS.